

In contrast, the market portfolio is *passive* and requires no trading. The action of rebalancing in equal weights imparts this strategy with a *rebalancing premium*. Rebalancing also turns out to be the foundation of an optimal long-run investing strategy. I cover these topics in the next chapter.

As you go down Table 3.16 from full-blown mean-variance to equal weights or the market portfolio, you estimate fewer parameters, and thus there are fewer things that can go wrong with the mean-variance optimization. The extreme cases are the equal weight or market weight positions, which require no analysis of data (except looking at market capitalizations in the case of the market portfolio).

Long-run means are very tricky to estimate. Sampling at weekly or daily frequencies does not allow you to more accurately estimate means—only extending the sample allows you to pin down the mean more precisely.³⁴ For any asset like the S&P 500, the only way to gauge the long-run return is to look at the index level at the beginning and at the end of the sample and divide it by time. It doesn't matter how it got to that final level; all that matters is the ending index value. Hence, we can only be more certain of the mean return if we lengthen time. This makes forecasting returns very difficult.³⁵ The minimum variance portfolio outperforms mean-variance because we remove all the errors associated with means.

Volatilities are much more predictable than means. High-frequency sampling allows you to estimate variances more accurately even though it does nothing for improving estimates of means. Higher frequency data also allows you to produce better estimates of correlations. But correlations can switch signs while variances can only be positive. Thus, variances are easier to estimate than correlations. Poor estimates of correlations also have severe effects on optimized mean-variance portfolios; small changes in correlations can produce big swings in portfolio weights.³⁶ Risk parity turns off the noise associated with estimating correlations. (More advanced versions of risk parity do take into account some correlation estimates.) In the horserace, risk parity produced higher Sharpe ratios (0.59 and 0.65 using variances and volatilities, respectively) than the minimum variance portfolio (which had a Sharpe ratio of 0.52).

In summary, the special cases of mean-variance perform better than the full mean-variance procedure because fewer things can go wrong with estimates of the inputs.

5.3. IMPLICATIONS FOR ASSET OWNERS

I went from the full mean-variance case to the various special cases in Table 3.16 by adding restrictions. To practice mean-variance investing, the investor should start at the bottom of Table 3.16 and begin from market weights. If you

³⁴ This is shown in a seminal paper by Merton (1980).

³⁵ I cover this further in chapter 8.

³⁶ Green and Hollifield (1992) provide bounds on the average correlation between asset returns that are required for portfolios to be well balanced.